

TYNSET-HANSMOEN, Norway

WMO: 012650

Lat: 62.269N

Lon: 10.732E

Elev: 1582

StdP: 13.87

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.3	-19.0	-30.9	1.0	-25.2	-24.0	1.5	-18.9	19.3	26.2	17.5	25.7	0.4	200	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	76.2	59.1	72.3	57.8	68.4	56.2	61.5	70.7	59.7	68.4	58.0	66.1	6.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.9	75.8	64.4	55.9	70.6	63.0	54.2	66.3	61.4	28.1	70.5	26.9	68.3	25.7	66.0	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	13.9	12.1	-30.2	80.7	6.8	4.0	-35.2	83.6	-39.2	85.9	-43.0	88.1	-48.0	91.0		
(5)			-30.7	64.9	7.1	1.4	-35.8	65.9	-39.9	66.7	-43.8	67.5	-49.0	68.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.7	13.1	16.2	23.8	34.8	43.6	51.2	56.3	53.5	46.5	35.2	25.0	15.8
	DBStd	17.81	14.66	12.98	11.29	6.60	7.21	6.04	5.39	5.05	5.88	7.52	10.90	14.62
	HDD50	6083	1144	947	812	458	227	57	9	22	136	461	751	1059
	HDD65	11072	1609	1367	1277	907	664	415	274	356	556	923	1200	1524
	CDD50	486	0	0	0	0	28	92	203	131	30	2	0	0
	CDD65	5	0	0	0	0	0	1	3	1	0	0	0	0
	CDH74	203	0	0	0	0	18	50	124	11	0	0	0	0
	CDH80	21	0	0	0	0	1	3	17	0	0	0	0	0
	WSAvg	4.6	4.2	4.5	5.1	5.1	5.4	5.2	4.6	4.1	4.5	4.3	3.9	3.9
	PrecAvg	24.4	1.6	1.4	1.4	1.2	1.8	2.9	3.4	3.3	2.2	1.9	1.7	1.7
(15) Precipitation	PrecMax	31.6	3.9	2.6	2.8	3.0	3.3	5.2	5.2	5.9	4.6	3.7	4.3	3.3
	PrecMin	19.4	0.1	0.2	0.4	0.4	0.7	0.8	1.5	1.0	0.9	0.7	0.4	0.9
	PrecStd	2.4	0.9	0.6	0.6	0.5	0.6	1.1	1.1	1.1	0.9	0.6	1.0	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	44.3	49.7	61.7	76.6	79.1	82.4	75.5	67.4	59.8	51.1	43.0
		MCWB	37.5	37.2	41.0	47.9	55.8	58.3	63.3	60.6	55.8	54.0	45.1	38.5
	2%	DB	38.3	39.9	45.1	56.1	69.8	74.4	77.6	71.3	63.5	52.9	45.0	40.1
		MCWB	34.7	35.4	38.2	44.8	53.7	57.0	60.8	58.0	54.3	47.1	41.8	36.1
	5%	DB	34.7	36.3	41.3	51.4	64.5	69.2	73.3	67.8	60.2	49.4	41.3	37.2
		MCWB	32.2	32.9	35.2	41.9	51.5	54.4	59.4	57.4	52.8	45.0	38.7	34.0
	10%	DB	32.2	33.3	38.4	47.3	58.5	64.6	69.1	64.1	57.2	46.4	37.7	34.4
		MCWB	30.5	30.7	33.8	39.9	48.5	53.4	57.5	56.2	51.3	43.2	35.5	32.0
	0.4%	WB	37.5	38.4	41.4	48.9	58.3	60.8	65.0	62.5	58.0	54.4	46.2	39.3
		MCDB	41.6	43.1	48.3	59.4	71.0	71.1	78.3	70.6	64.6	58.7	50.6	43.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.0	35.7	38.6	45.7	55.3	58.6	62.6	60.5	55.5	48.1	42.3	36.3
		MCDB	38.0	39.5	44.6	54.2	67.2	70.5	73.1	67.6	61.3	51.7	45.5	39.4
	5%	WB	32.4	32.9	36.1	42.9	52.4	56.6	60.7	58.9	53.6	45.6	38.9	34.1
		MCDB	34.4	36.1	40.3	50.4	62.4	67.3	70.4	65.3	59.0	48.7	41.2	36.7
	10%	WB	30.4	30.9	34.0	40.0	49.4	54.5	58.7	57.1	51.7	43.2	35.7	32.1
		MCDB	32.4	33.3	37.9	46.7	58.4	63.0	67.4	63.2	56.4	46.5	37.5	34.2
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	17.6	21.8	19.5	20.0	20.2	19.8	18.9	18.0	14.0	11.9	15.5
		MCDBR	15.3	17.1	20.8	27.3	32.0	31.0	29.3	25.5	23.2	17.8	14.0	17.7
		MCWBR	12.8	13.3	14.6	16.9	17.1	15.4	14.6	14.1	14.7	13.2	11.1	14.8
	5% WB	MCDBR	14.9	16.3	19.2	25.2	28.0	27.0	26.0	20.8	20.3	15.8	13.3	17.1
		MCWBR	12.6	12.9	13.8	16.1	15.4	14.1	13.7	12.4	13.4	12.1	10.8	14.5
(40) Clear-Sky Solar Irradiance	taub		0.198	0.251	0.283	0.313	0.327	0.325	0.338	0.328	0.297	0.279	0.210	0.170
	taud		2.085	2.260	2.251	2.281	2.433	2.493	2.469	2.488	2.555	2.472	2.146	1.975
	Ebn at Noon		164	232	264	278	283	286	278	272	263	223	154	108
	Edh at Noon		10	19	28	33	31	30	30	27	21	16	9	6
(44) All-Sky Solar Radiation	RadAvg		78	289	774	1359	1550	1580	1471	1136	721	322	101	41
	RadStd		10	39	63	95	101	139	161	106	73	43	16	3

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-216	-259	N/A	N/A

Nomenclature: See separate page